DEVELOPER PRODUCTIVITY PATTERN ANALYZER

Full-Stack Application with React Frontend, Flask Backend, and Machine Learning

PROJECT OVERVIEW

The Developer Productivity Pattern Analyzer is a full-stack application designed to uncover meaningful productivity patterns using data-driven machine learning. It brings together a React frontend, Flask backend, database layer, and ML pipeline in a single interactive system for uploading data, running analysis, and exploring results. By using clustering to group similar productivity behaviors and anomaly detection to highlight unusual trends, the platform delivers actionable insights from developer activity data. Interactive visualizations help teams and stakeholders quickly understand findings and support better decision-making.

TECHNICAL STACK - FRONTEND

Built with React 18+ and powered by the Vite development server on port 3000, the frontend delivers fast startup, hot module replacement, and an efficient development experience for building responsive interfaces.

Modern CSS animations create a polished user experience, while the application includes a user-friendly data upload interface and dynamic dashboard visuals for exploring productivity insights interactively.

BACKEND & DATABASE

Python 3.8+ powers a Flask REST API running on port 8000, connecting the frontend to backend services. Core settings are managed through backend/config.py for clean, centralized configuration.

MongoDB stores developer records at mongodb://localhost:27017, supporting reliable access during analysis. Seeding scripts populate test data and generate sample CSV files for quick setup and testing.

MACHINE LEARNING

- K-Means Clustering groups developers into productivity patterns based on performance data.
- Isolation Forest identifies unusual records and highlights anomalies in developer activity.
- Both models are integrated into the Flask backend processing pipeline after data ingestion.
- Results are sent to the frontend dashboard for clear visualization through charts and insights.

LANDING & UPLOAD FEATURES

Modern landing page with smooth CSS animations and a clean interface that welcomes users into the analysis workflow.

User-friendly CSV upload functionality makes it easy to import developer data and start analysis in just a few clicks.

Includes the sample_developers.csv test file so users can quickly explore the app without preparing their own dataset.

PRODUCTION ANALYSIS & DASHBOARD

Run Production Analysis to trigger the backend machine learning pipeline and process uploaded developer data.

Interactive dashboard presents clustering distribution charts and anomaly detection visuals for rapid pattern discovery.

Detailed DataTable enables record-level inspection, metric comparison, and deeper developer productivity insights.

BACKEND SETUP PROCESS

1. Create and activate a Python virtual environment.
2. Install backend dependencies from requirements.txt.

1. Run seed_db.py to seed MongoDB with test data.
2. Start the Flask API server on port 8000.

FRONTEND SETUP PROCESS

1. Open a terminal and navigate to the frontend folder.
2. Install the required Node.js dependencies with npm install.

1. Start the Vite development server with npm run dev.
2. Access the frontend in your browser at http://localhost:3000.

APPLICATION WORKFLOW

- Start on the landing page and click "Get Started" to move into the analysis workflow.
- On the Upload Data page, add the sample_developers.csv file and press "Run Production Analysis" to trigger backend ML processing.
- The app redirects to the Dashboard, where users review clustering and anomaly outputs, interactive charts, and detailed developer data tables.

THANK YOU!

Thank you for your attention! Questions welcome.